

**Project Name: ERP Analytics Agent using Microsoft Copilot Studio**  
**Supervisor: Dr.Gopakumar**  
**Date: 6/24/2016**  
**Company: Merp Systems Private Limited**

## **1. Executive Summary**

The project is a RAG agent that analyzes data from a serverless SQL Server to create financial reports. The aim was to automate the reports that ERP managers often create while exploring the technologies available in Microsoft Copilot Studio. Particularly, the use of the new MCP server feature and the ability to use natural language to query a SQL Server that is not on-premises.

## **3. Inspiration**

A client of Merp Systems in Oman is a company that stores cell phone records of all calls made within the country as part of the government's data collection program. This data is stored on many on-premises servers, but the government did not consent to migrating this data to the cloud due to security concerns. The idea for this project was presented as if the government consented to this. Would it be possible to use Power Platform, Copilot Studio, and other Microsoft resources to create an MCP server that could reliably query this data? MCP servers are a newer feature added to Copilot Studio, and my Project Manager at Merp Systems, Mr. Gopakumar, wanted to experiment with this feature and see its capabilities.

He was also interested in D365 F&O and wanted to see if it was possible to query F&O data using the MCP server. F&O is Microsoft's ERP solution, which is very secure and uses complex relational databases to store information for a client's company. I was given access to a trail environment called USMF for Contonso Ent., which is a fictional multinational supply and delivery company. After some experimenting, we found that the security on this application is tight, the schema is not readily available and complex, and even with the data I was able to secure, the results were not accurate since it was a test environment that multiple other employees had used for training how to use F&O, and now contained many false and illogical values.

After discussion with my Project Manager, it was decided that an agent that uses MCP to query any ERP system and automate report writing would be the final project idea.

## **4. Existing Systems & Why This Is Different**

Existing ERP systems like D365 F&O contain large quantities of tables with complex relations that are not accessible to the public. Querying it using only OData requires extensive knowledge of the connections between tables, and Copilot Studio has no built-in connector to this data. Financial data in most ERP systems is hard to extract due to tight security systems and limited visibility of the architecture. This system is a simple version of those complex systems to see if creating an agent to query this data is viable.

## 5. Functional Requirements

1. Create an agent customized to provide financial help
2. Use an MCP server to Query that data
3. Store the data in the cloud instead of on-premises SQL server
4. Generate downloadable reports that analyze the output from the MCP server
5. Avoid black box agents and include explanations and calculations used in the output
6. Convert Natural Language to the appropriate SQL query
7. Allow anyone with access to the ERP software to use the chatbot without additional sign-ins

## 6. Non-Functional Requirements

1. Model an ERP system and how a chatbot would appear to the employee
2. Format reports so they are readable and easy to analyze
3. Save space by not permanently joining any tables to create a view; instead, only join tables when needed

## 7. Data Model & Schema Overview

Adventure Works Cycles is a fictitious, large-scale, multinational manufacturing company created by Microsoft. It produces and distributes metal and composite bicycles, along with related parts and accessories, to commercial markets across North America, Europe, and Asia. It contains information from 2010-2013. It is provided by Microsoft to train SQL skills and understanding of relational databases, so the numbers are not realistic. It's ERP functions across five core departments: Human Resources, Sales, Production, Purchasing, and People.

This is its schema:

- DimAccount(AccountKey PK, ParentAccountKey, AccountDescription, AccountType)
- DimOrganization(OrganizationKey PK, ParentOrganizationKey, OrganizationName, CurrencyKey)
- DimDate(DateKey PK, CalendarYear, MonthNumberOfYear)
- DimDepartmentGroup(DepartmentGroupKey PK, ParentDepartmentGroupKey, DepartmentGroupName)
- DimScenario(ScenarioKey PK, ScenarioName)

RELATIONSHIPS (FactFinance)

- AccountKey → DimAccount.AccountKey

- OrganizationKey → DimOrganization.OrganizationKey
- DateKey → DimDate.DateKey
- DepartmentGroupKey → DimDepartmentGroup.DepartmentGroupKey
- ScenarioKey → DimScenario.ScenarioKey

#### HIERARCHY (SELF-REF)

- DimAccount.ParentAccountKey → DimAccount.AccountKey
- DimOrganization.ParentOrganizationKey → DimOrganization.OrganizationKey
- DimDepartmentGroup.ParentDepartmentGroupKey → self
- DimEmployee.ParentEmployeeKey → self

#### ADDITIONAL CORE JOINS (COMMON DW)

- DimCustomer.GeographyKey → DimGeography
- DimGeography.SalesTerritoryKey → DimSalesTerritory
- DimProduct.ProductSubcategoryKey → DimProductSubcategory → DimProductCategory

#### SALES FACT DATE RULE

- FactInternetSales / ResellerSales:

OrderDateKey, DueDateKey, ShipDateKey → DimDate.DateKey

(default = OrderDateKey)

#### PRODUCT CHAIN

- DimProduct → DimProductSubcategory → DimProductCategory

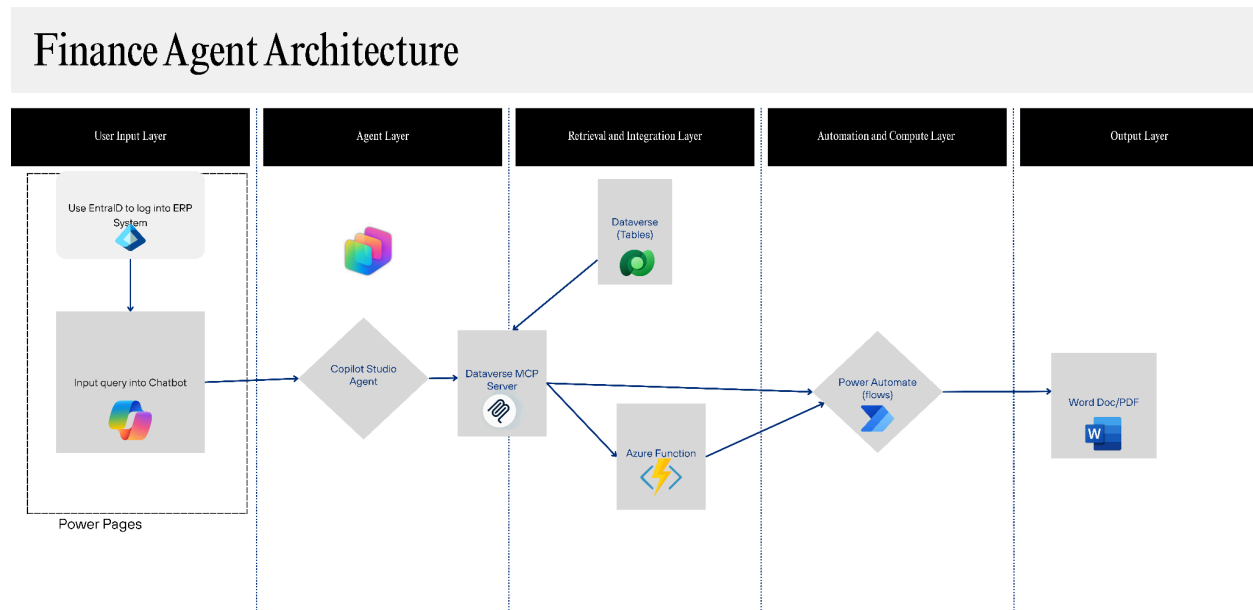
#### LEAF TABLES (no outgoing FK)

- DimCurrency, DimPromotion, DimSalesReason, DimSalesTerritory, DimScenario

These relationships contain a lot of parent-child relationships where only the parents have data, and the child to whom it connects has empty rows. Due to taking data from a data warehouse, a fan trap occurred where fact tables stored data at multiple levels of hierarchy simultaneously, resulting in silent errors but with inflated numbers due to double counting. Departments located on multiple levels and in multiple countries had larger variance percentages than departments

only located in one country, which is how this error was found. Example: North American operations-> northwest division-> northwest district. Adding the above schema to the agent as knowledge solved this problem as the deepest level of the hierarchy could be identified and only information from that was taken for analysis.

## 8. Architecture Diagram



## 9. Technical Approach

## 10. Design Decisions Log

1. Chose to store in Dataverse instead of other types of cloud storage like Azure Databricks due to Copilot Studio directly being connected to it and wanting to test the Dataverse MCP server.
2. Chose to add an Azure function to expand the existing functionality of the MCP server. Creating a custom MCP server with dataverse MCP server functions, as well as custom functions for joins and subqueries, was considered, but the idea was rejected as it would introduce unnecessary complexity. Instead, logically connecting the Azure function to execute after the MCP server gave the same result as a custom MCP server would have.
3. Presenting the data in PDF form was considered, but it required extra paid services. Output as an Excel file was also considered, but it would not seem like a professional report. Instead, a Word document was selected.
4. Originally, the documents were to be stored in OneDrive, but that requires OAuth, which means only the creator of the agent would have access to the reports. SharePoint was used instead with an enterprise account so that any employee can access the saved reports.
5. Since the agent was to be added to an ERP System, it had to be accessible to anyone who has access to that ERP system. To do this, an iframe of the Copilot Studio agent

was used, but because of this decision, all OAuth security checks and connections were reconfigured. This was deemed safe since ERP managers are the only ones with access to an ERP system, and they already have access to all the data that the agent has access to.

6. Creating a URL with blob storage was chosen as it has no storage cost and expires in 24 hours, so the cloud storage limits is optimized.
7. In Azure Function, both Web Api or TDS endpoint could have been used to connect to Dataverse and Copilot Studio, but a TDS endpoint was chosen since full T-SQL support was the priority, even if the setup was more complex
8. Intellisense was considered for auto-checking the Microsoft SQL Server Management Studio, but was not used due to unreliable caching and a tendency to get column names wrong
9. Including an orchestration agent or separate child agents for each type of report was considered, but the MCP server fulfilled the role of an orchestration agent by deciding which tools should be used. Custom prompts were used for each type of report instead. This helps avoid overcomplication and the chances of a timeout.
10. Using Slot filling technique with word templates to create properly formatted reports was the initial plan, but it required large amounts of templates and specific Power Automate flows for each. Though the current HTML tables in the document are simple, they are easy to read, don't affect the comprehension of the reports, and use much fewer tokens.

## **11. Services & Tools Used**

- a. Adventureworks Datawarehouse
  - i. A sample database created by Microsoft is meant to help users learn to use Microsoft SQL Management Studio and learn SQL queries by simulating a multinational bicycle parts manufacturing and distribution company
- b. Microsoft SQL Server Management Studio
  - i. Used to download the AdventureWorks Data Warehouse SQL server and access what data is available in the server. It was used to analyze the relational database architecture to understand what joins could be made.
  - ii. The main idea was to make all tables available in this server that are on-premises to a cloud storage while maintaining the relational database schema
- c. Microsoft Dataverse
  - i. A licensed service where the tables from the SQL server were copied and labeled according to primary and foreign keys to maintain the relational database structure

Power Apps

Search

Environment  
Tony Stark's Environment

aw

| Table                           | Name                 | Type     | Managed | Customized | Customizable | Tags   |
|---------------------------------|----------------------|----------|---------|------------|--------------|--------|
| AdventureWorksDWBuildVersion-aw | crfaa_adventure...   | Standard | No      | Yes        | Yes          | Custom |
| DatabaseLog-aw                  | crfaa_databaselo...  | Standard | No      | Yes        | Yes          | Custom |
| DimAccount-aw                   | crfaa_dimaccoun...   | Standard | No      | Yes        | Yes          | Custom |
| DimCurrency-aw                  | crfaa_dimcurrenc...  | Standard | No      | Yes        | Yes          | Custom |
| DimCustomer-aw                  | crfaa_dimcustom...   | Standard | No      | Yes        | Yes          | Custom |
| DimDate-aw                      | crfaa_dimdateaw      | Standard | No      | Yes        | Yes          | Custom |
| DimDepartmentGroup-aw           | crfaa_dimdepart...   | Standard | No      | Yes        | Yes          | Custom |
| DimEmployee-aw                  | crfaa_dimemploy...   | Standard | No      | Yes        | Yes          | Custom |
| DimGeography-aw                 | crfaa_dimgeogra...   | Standard | No      | Yes        | Yes          | Custom |
| DimOrganization-aw              | crfaa_dimorganiz...  | Standard | No      | Yes        | Yes          | Custom |
| DimProduct-aw                   | crfaa_dimproduc...   | Standard | No      | Yes        | Yes          | Custom |
| DimProductCategory-aw           | crfaa_dimproduc...   | Standard | No      | Yes        | Yes          | Custom |
| DimProductSubcategory-aw        | crfaa_dimproduc...   | Standard | No      | Yes        | Yes          | Custom |
| DimPromotion-aw                 | crfaa_dimpromot...   | Standard | No      | Yes        | Yes          | Custom |
| DimReseller-aw                  | crfaa_dimresell...   | Standard | No      | Yes        | Yes          | Custom |
| DimSalesReason-aw               | crfaa_dimsalesre...  | Standard | No      | Yes        | Yes          | Custom |
| DimSalesTerritory-aw            | crfaa_dimsalester... | Standard | No      | Yes        | Yes          | Custom |
| DimScenario-aw                  | crfaa_dimscenari...  | Standard | No      | Yes        | Yes          | Custom |

Power Apps

Search

Environment  
Tony Stark's Environment

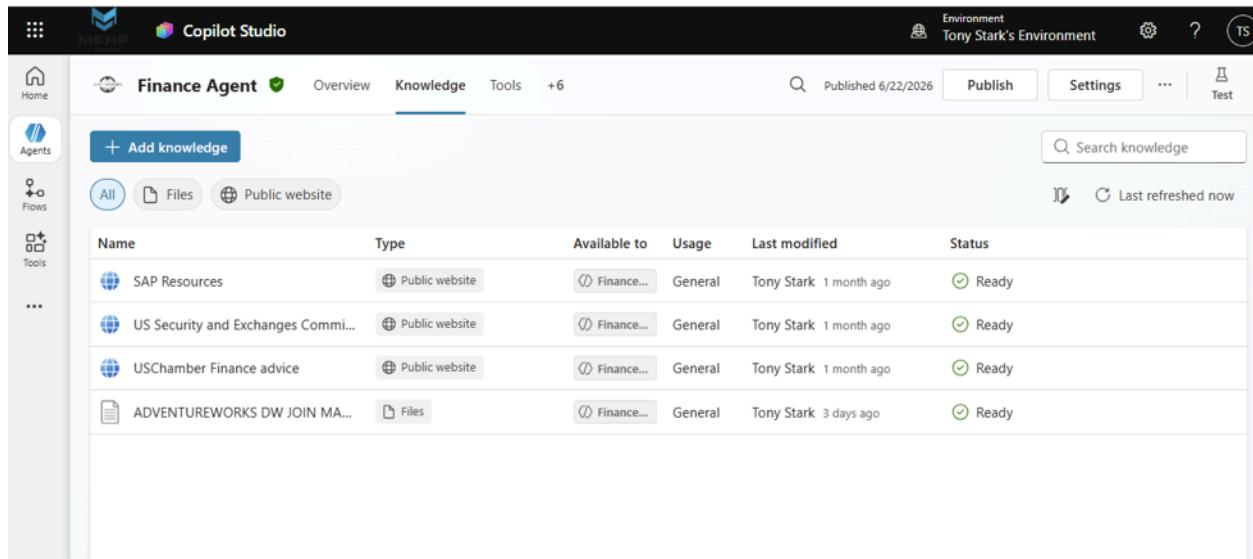
aw

|                                          |                       |          |    |     |     |        |
|------------------------------------------|-----------------------|----------|----|-----|-----|--------|
| DimScenario-aw                           | crfaa_dimscenari...   | Standard | No | Yes | Yes | Custom |
| FactAdditionalInterProductDescription... | crfaa_factadditio...  | Standard | No | Yes | Yes | Custom |
| FactCallCenter-aw                        | crfaa_factcallcent... | Standard | No | Yes | Yes | Custom |
| FactCurrencyRate-aw                      | crfaa_factcurrenc...  | Standard | No | Yes | Yes | Custom |
| FactFinance-aw                           | crfaa_factfinance...  | Standard | No | Yes | Yes | Custom |
| FactInternetSales-aw                     | crfaa_factinterne...  | Standard | No | Yes | Yes | Custom |
| FactInternetSalesReason-aw               | crfaa_factinterne...  | Standard | No | Yes | Yes | Custom |
| FactProductInventory-aw                  | crfaa_factproduc...   | Standard | No | Yes | Yes | Custom |
| FactResellerSales-aw                     | crfaa_factresell...   | Standard | No | Yes | Yes | Custom |
| FactSalesQuota-aw                        | crfaa_factsalesqu...  | Standard | No | Yes | Yes | Custom |
| FactSurveyResponse-aw                    | crfaa_factsurveyr...  | Standard | No | Yes | Yes | Custom |
| NewFactCurrencyRate-aw                   | crfaa_newfactcur...   | Standard | No | Yes | Yes | Custom |
| ProspectiveBuyer-aw                      | crfaa_prospectiv...   | Standard | No | Yes | Yes | Custom |
| sysdiagrams-aw                           | crfaa_sysdiagram...   | Standard | No | Yes | Yes | Custom |
| vAssocSeqLineItems-aw                    | crfaa_vassocseqli...  | Standard | No | Yes | Yes | Custom |
| vAssocSeqOrders-aw                       | crfaa_vassocseqo...   | Standard | No | Yes | Yes | Custom |
| vDMPrep-aw                               | crfaa_vdmprepaw       | Standard | No | Yes | Yes | Custom |
| vTargetMail-aw                           | crfaa_vtargetmail...  | Standard | No | Yes | Yes | Custom |
| vTimeSeries-aw                           | crfaa_vtimeseries...  | Standard | No | Yes | Yes | Custom |

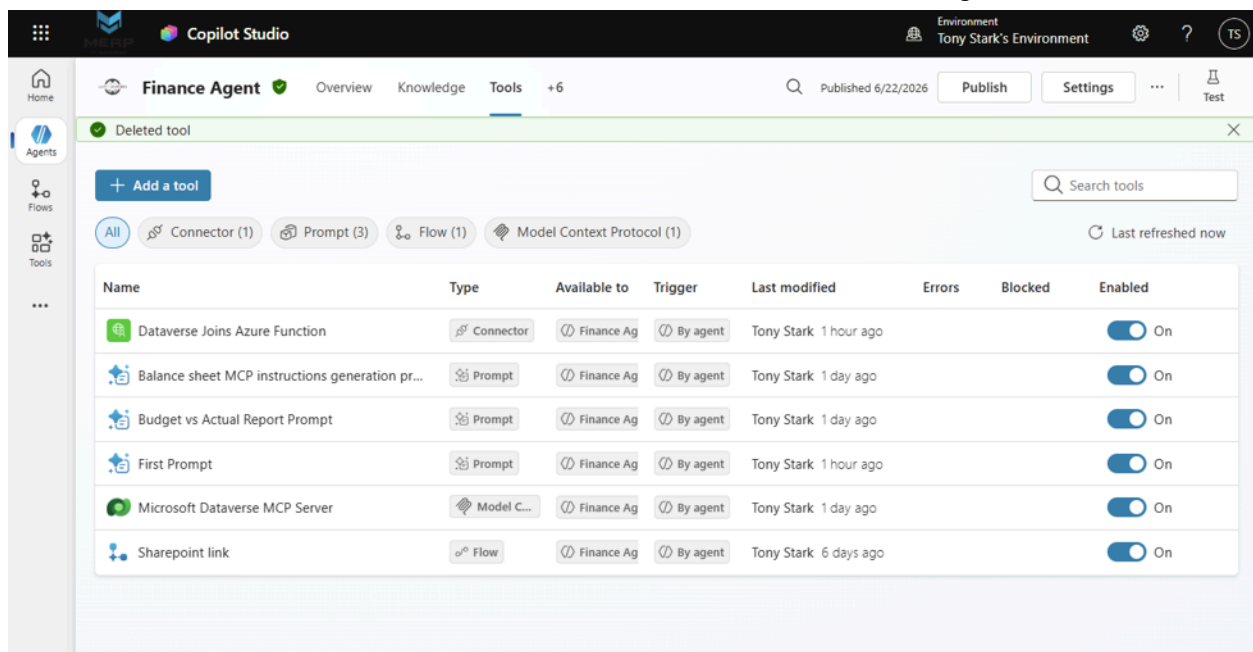
Screenshot shows all the tables stored in Dataverse

d. Copilot Studio Agent

- i. Microsoft licensed tool that allows the creation of agents by adding instructions, knowledge, tools, etc.



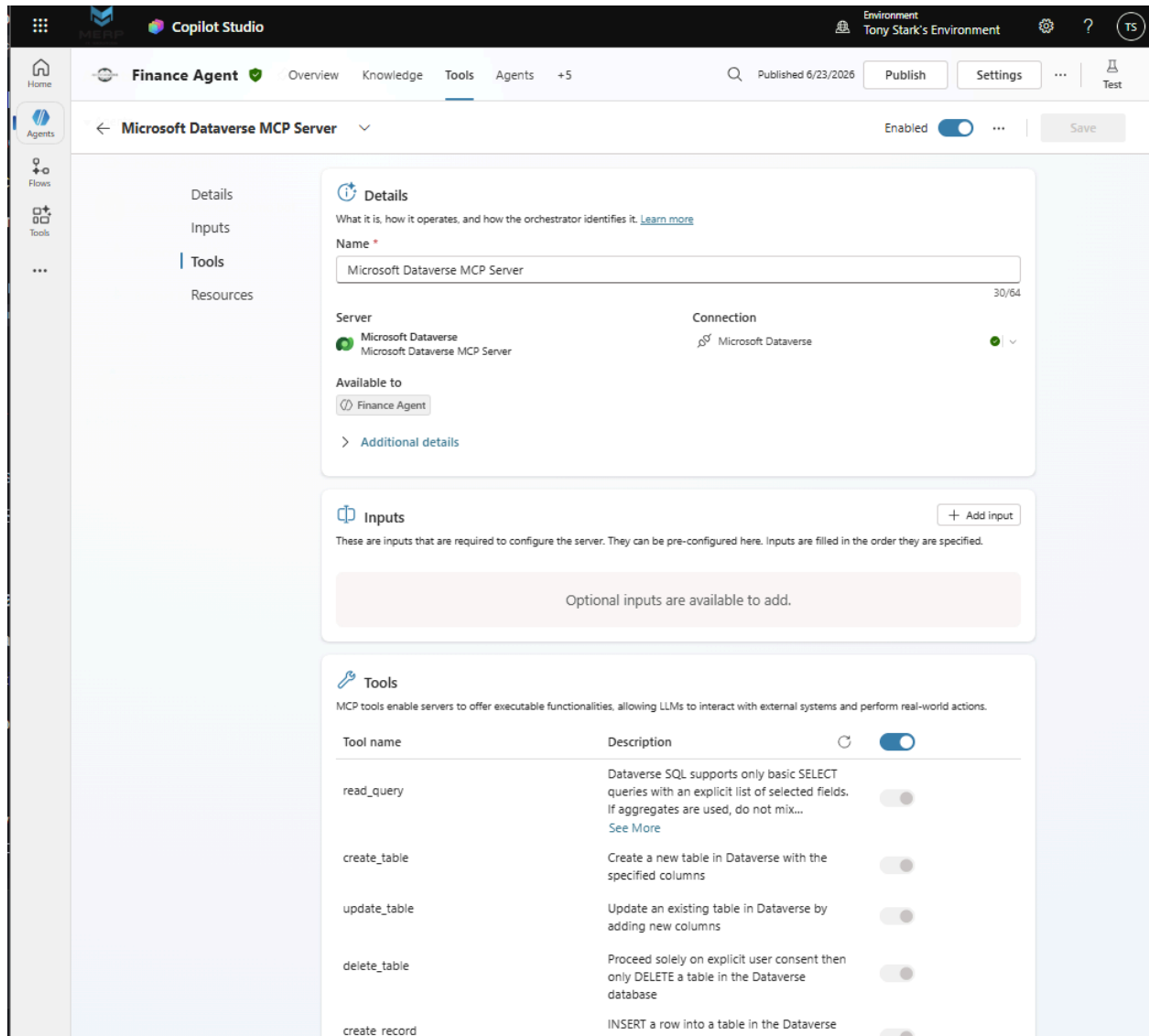
*Database schema and financial advice resources were added as knowledge*



*Tools that the agent can use, some were custom made while others are readily available in Copilot Studio*

- e. Microsoft Dataserve MCP Server
  - i. A newer feature in Copilot Studio that has multiple functions/ tools inside it, each used to extract information from Microsoft Dataserve. It acts as an orchestration agent and chooses which of these tools should be used and when
  - ii. Tools include read query, search data, create record, update record, delete record, create table, delete table, describe, file upload, file download

- iii. It lacked the ability to perform complex joins between tables or create subqueries, but was useful in deciding which tables and columns were most relevant to the user prompt.
- iv. Since Dataverse MCP can technically access all tables available tables in Dataverse, a custom application user was created that can only access the AdventureWorks tables



*Dataverse MCP Server in Copilot Studio*

For the best experience viewing this content, open it in a specific application. Choose an app

← Show Chart + New Delete Refresh Email a Link Flow Run Report Excel Templates Share

Active Allowed MCP Clients Edit columns Edit filters Filter by keyword

| <input type="checkbox"/> Name ↑                       | Unique Name                   | Application Id                        | Is Enabled | Created On         |
|-------------------------------------------------------|-------------------------------|---------------------------------------|------------|--------------------|
| <input type="checkbox"/> Agent 365 Tools              | agent365tools                 | ea9ffc3e-8a23-4a7d-836d-234d7c756...  | Yes        | 5/28/2026 4:24 PM  |
| <input type="checkbox"/> Copilot Studio Custom MCP v1 | new_copilotstudiocustommcp... | a825b355-8b59-4cf3-a7ea-b6ce89c6c...  | Yes        | 6/1/2026 5:31 AM   |
| <input type="checkbox"/> Copilot Studio Custom MCP v2 | new_copilotstudiocustommcp... | a825b355-8b59-4cf3-a7ea-b6ce89c6c...  | Yes        | 6/4/2026 11:27 AM  |
| <input type="checkbox"/> Copilot Studio Custom MCP v3 | new_copilotstudiocustommcp... | a825b355-8b59-4cf3-a7ea-b6ce89c6c...  | Yes        | 6/10/2026 11:06 AM |
| <input type="checkbox"/> Dataverse MCP CLI tool       | dataversemcpcli               | 0c412cc3-0dd6-449b-987f-05b053db...   | Yes        | 5/2/2026 1:43 PM   |
| <input type="checkbox"/> Microsoft Copilot Studio App | microsoftcopilotstudio        | 7ab7862c-4c57-491e-8a45-d52a7e02...   | Yes        | 5/2/2026 1:43 PM   |
| <input type="checkbox"/> Microsoft Cowork             | microsoftcowork               | 6ab48b67-cd74-4ad4-81af-59329845...   | No         | 5/28/2026 4:24 PM  |
| <input type="checkbox"/> Microsoft GitHub Copilot App | microsoftgithubcopilot        | aebc6443-996d-45c2-90f0-388ff96faa... | Yes        | 5/2/2026 1:43 PM   |
| <input type="checkbox"/> vibepowerapps                | developeragentsprod           | 8e38132a-0fcc-47e2-8178-724bda6fc...  | Yes        | 5/28/2026 4:24 PM  |

## MCP server client permission customization

### f. Azure Function App

- To add more functions/ tools to the MCP server, an Azure function was created in Python and uploaded to the Azure function app. It uses an HTTP trigger and sends information back to the Copilot Studio agent via an HTTP trigger
- It was needed to do multitable joins, read more lines since the MCP server caps at 1000 records, perform subqueries, and window functions
- To avoid OAuth, this was done with the combination of tenant id, client id, client secret, Dataverse instance URL, and environment details. These were created by an Azure function app called copilot-dataverse-app, and the function code was developed in VS Code before being uploaded to the cloud using the func-aw-query resource group. There, the use of the Azure function can be monitored, and the code is maintained off-premises. A TDS endpoint was used to query dataverse
- The azure function code can be found it github:  
<https://github.com/Rayachan/adventureworks-ai-agent/tree/main>

Home > func-aw-query

**func-aw-query\_group** Resource group

Export resource groups using Bicep or Terraform | Generate Bicep code to duplicate this resource group. +1

Search

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

Subscription (move): [Microsoft Partner Network](#)

Subscription ID: 4f5c3771-7079-4963-9240-3e7a4f68c0e2

Deployments: 3 Succeeded

Location: Canada Central

Tags (edit): Add tags

Resources Recommendations

Filter for any field...

Type equals all Location equals all Add filter

| Name                              | Type                      | Location       |
|-----------------------------------|---------------------------|----------------|
| ASP-funcawquerygroup-8143         | App Service plan          | Canada Central |
| Failure Anomalies - func-aw-query | Smart detector alert rule | Global         |
| func-aw-query                     | Function App              | Canada Central |
| func-aw-query                     | Application Insights      | Canada Central |
| funcawquerygroup87a1              | Storage account           | Canada Central |

### Resource group containing the function app func-aw-query

Home > func-aw-query > func-aw-query\_group

**func-aw-query** Function App

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

AI (preview)

Resource visualizer

Functions

Deployment

Settings

Performance

Development Tools

API

Monitoring

Automation

Support + troubleshooting

Essentials

Resource group (move): [func-aw-query\\_group](#)

Status: Running

Location (move): Canada Central

Subscription (move): [Microsoft Partner Network](#)

Subscription ID: 4f5c3771-7079-4963-9240-3e7a4f68c0e2

Tags (edit): Add tags

Functions Properties Notifications (0)

Set up local environment Refresh

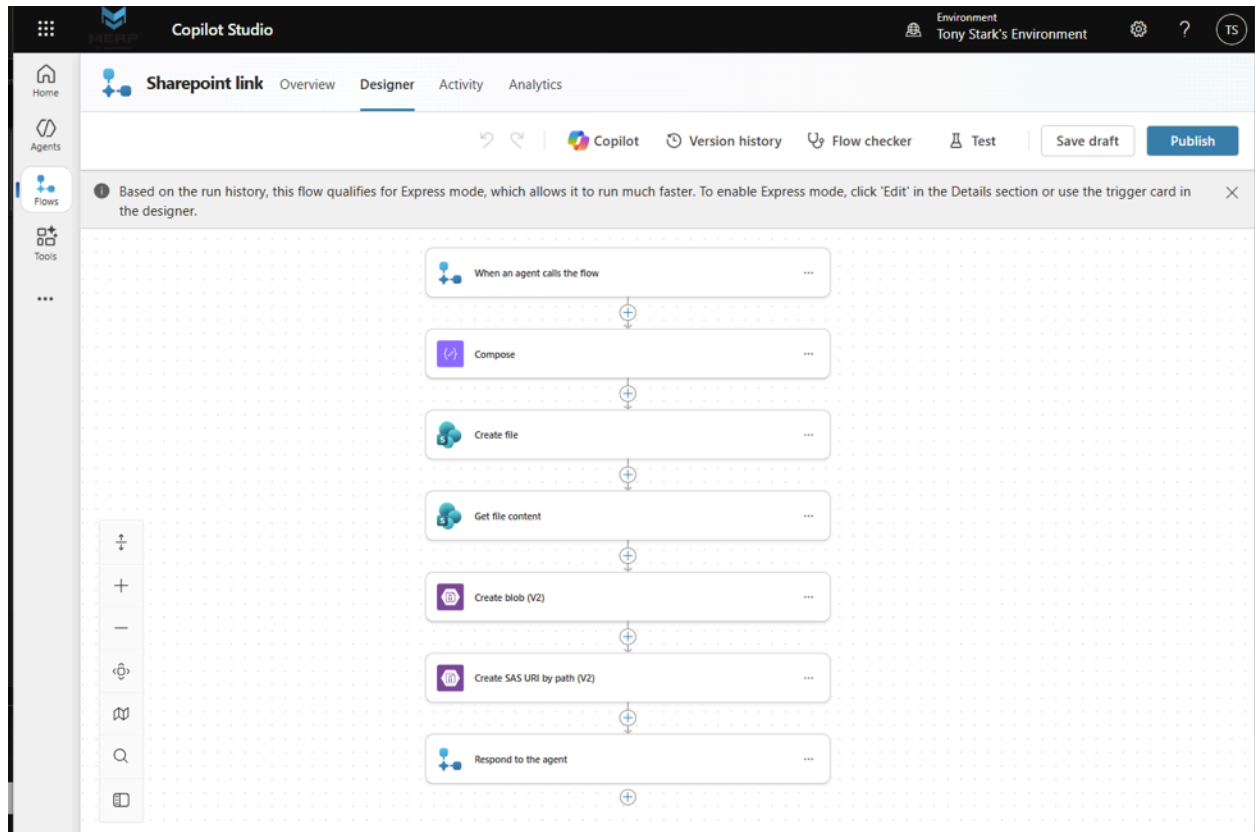
Filter by name...

| Name            | Trigger | Status  | Monitor              |
|-----------------|---------|---------|----------------------|
| query_dataverse | HTTP    | Enabled | Invocations and more |

Azure Function App that has query\_dataverse, which is the Azure Function uploaded to the cloud

#### g. Microsoft Power Automate

- i. A Microsoft License service that allows the creation of flows that the agent uses to perform a series of actions. Power Automate flow was used to take the information returned from the MCP server and Azure function and format it into a document by converting JSON to HTML and then formatting it into a Word document that is stored in Microsoft SharePoint. Then, Azure blob storage was used to generate a link to this document



*The agent calls this flow when it needs the report as a document*

h. Azure Blob Storage

- i. A cost-efficient location to store files that is also able to create links to the reports stored within. It is connected to a resource group called aw-reports-rg with a storage account within it called financeagentblob storage. Inside this blob storage, a container was created to store all these reports

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

ironman@merpsystems...  
MERP SYSTEMS, INC. (MERPSYST...

Home

adventureworks-reports

Container

Search

+ Add Directory ↑ Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

adventureworks-reports

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 70 items

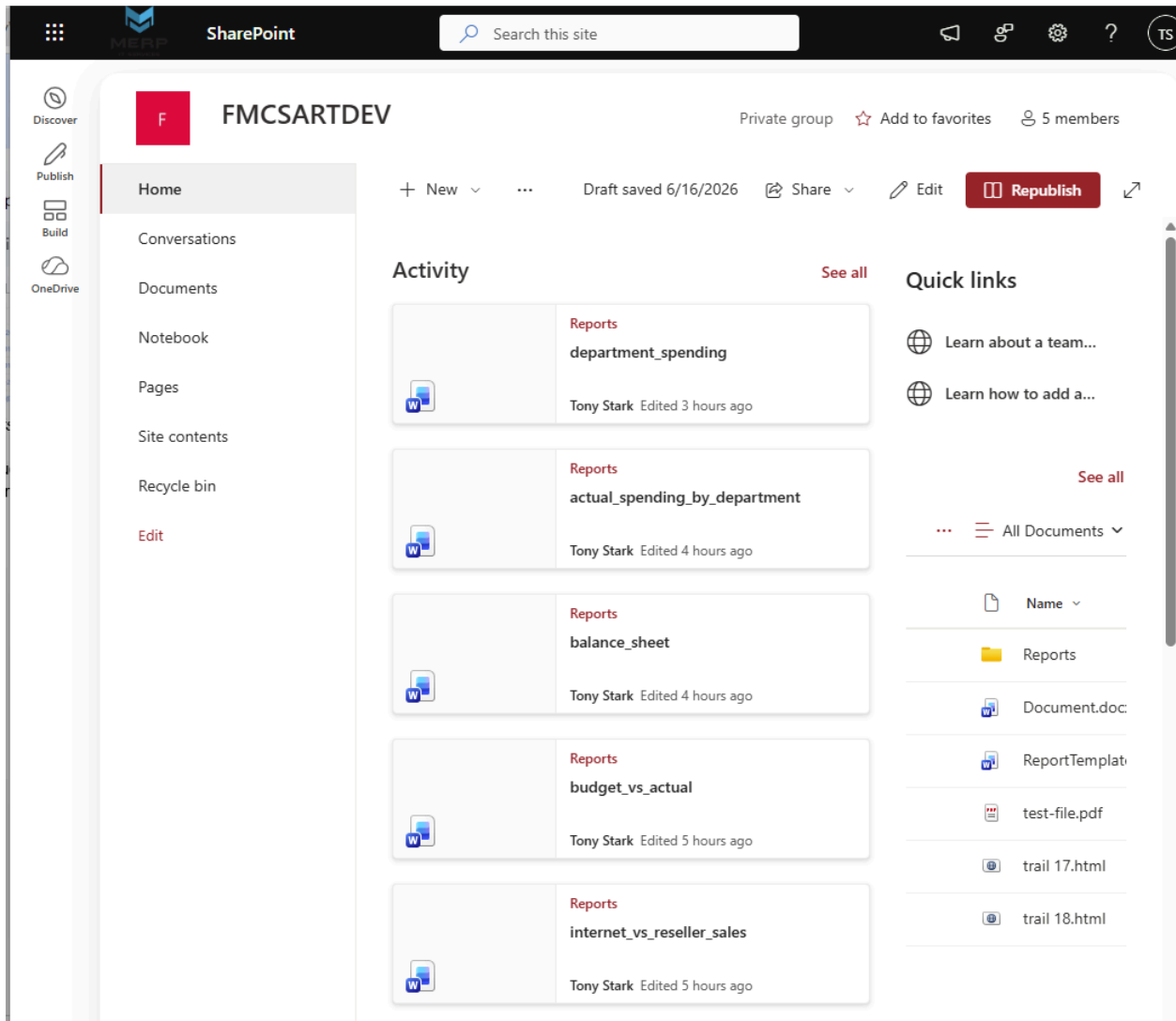
| <input type="checkbox"/> | Name                                    | Last modified         | Access tier    | Blob type  | Size       | Lease state |
|--------------------------|-----------------------------------------|-----------------------|----------------|------------|------------|-------------|
| <input type="checkbox"/> | balance_sheet-<table border="1" ce...   |                       |                |            |            | ...         |
| <input type="checkbox"/> | -20260608-0706.html                     | 6/8/2026, 12:36:46 PM | Hot (Inferred) | Block blob | 400 B      | Available   |
| <input type="checkbox"/> | -20260608-0710.html                     | 6/8/2026, 12:40:07 PM | Hot (Inferred) | Block blob | 400 B      | Available   |
| <input type="checkbox"/> | -20260608-0839.html                     | 6/8/2026, 2:09:55 PM  | Hot (Inferred) | Block blob | 1011 B     | Available   |
| <input type="checkbox"/> | -20260608-0841.html                     | 6/8/2026, 2:11:42 PM  | Hot (Inferred) | Block blob | 1000 B     | Available   |
| <input type="checkbox"/> | -20260608-0853.html                     | 6/8/2026, 2:23:16 PM  | Hot (Inferred) | Block blob | 424 B      | Available   |
| <input type="checkbox"/> | -20260608-0905.html                     | 6/8/2026, 2:35:06 PM  | Hot (Inferred) | Block blob | 21 KiB     | Available   |
| <input type="checkbox"/> | -20260608-0908.html                     | 6/8/2026, 2:38:04 PM  | Hot (Inferred) | Block blob | 37.41 KiB  | Available   |
| <input type="checkbox"/> | -20260608-0910.html                     | 6/8/2026, 2:41:01 PM  | Hot (Inferred) | Block blob | 37.41 KiB  | Available   |
| <input type="checkbox"/> | -20260608-0921.html                     | 6/8/2026, 2:51:26 PM  | Hot (Inferred) | Block blob | 21.26 KiB  | Available   |
| <input type="checkbox"/> | -20260608-1007.pdf                      | 6/8/2026, 3:37:57 PM  | Hot (Inferred) | Block blob | 21.55 KiB  | Available   |
| <input type="checkbox"/> | -20260608-1010.pdf                      | 6/8/2026, 3:40:30 PM  | Hot (Inferred) | Block blob | 43.27 KiB  | Available   |
| <input type="checkbox"/> | Balance Sheet - Actual 2011-202606...   | 6/9/2026, 4:00:53 PM  | Hot (Inferred) | Block blob | 37.4 KiB   | Available   |
| <input type="checkbox"/> | Balance Sheet - Fiscal Year 2010-202... | 6/9/2026, 3:37:49 PM  | Hot (Inferred) | Block blob | 51.22 KiB  | Available   |
| <input type="checkbox"/> | Balance Sheet - Fiscal Year 2010-202... | 6/9/2026, 3:51:11 PM  | Hot (Inferred) | Block blob | 51.23 KiB  | Available   |
| <input type="checkbox"/> | Balance Sheet Report - 2011-202606...   | 6/9/2026, 10:16:06 AM | Hot (Inferred) | Block blob | 37.84 KiB  | Available   |
| <input type="checkbox"/> | Balance Sheet Report - 2012 (Detail...  | 6/9/2026, 2:36:24 PM  | Hot (Inferred) | Block blob | 147.79 KiB | Available   |

Add or remove favorites by pressing Ctrl+Shift+F

*Inside the container are all the reports that were generated with their details*

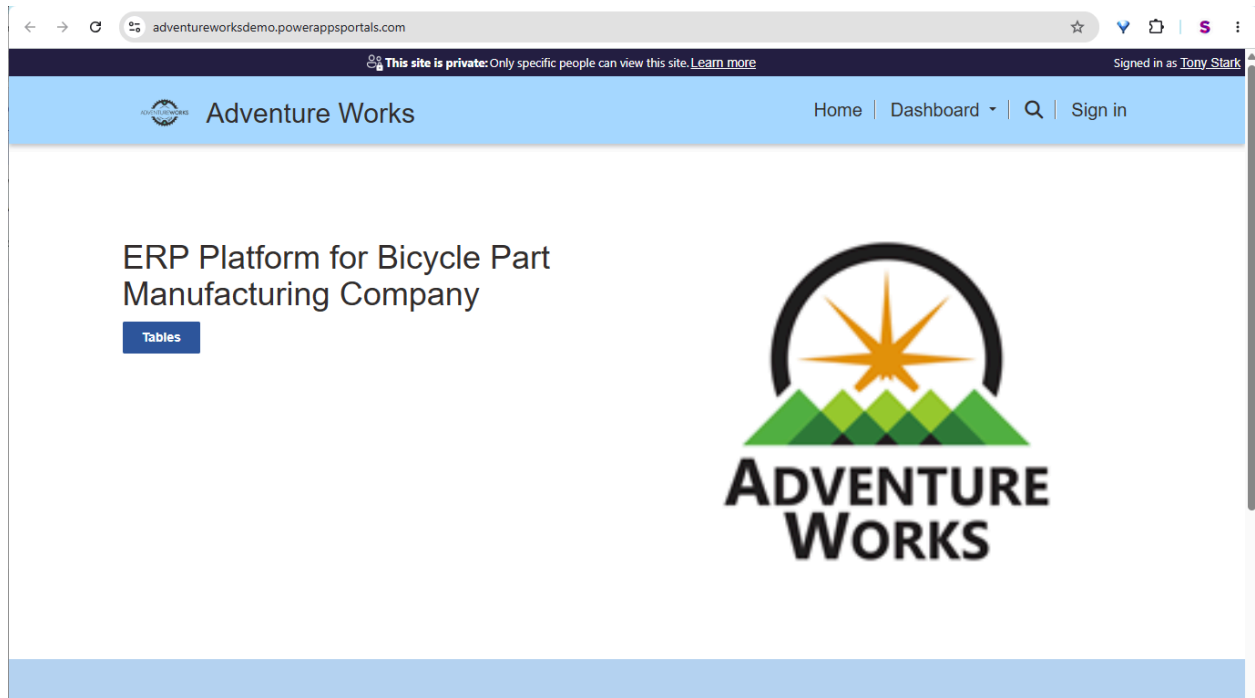
i. Microsoft Sharepoint

i. Another Microsoft cloud-based document management platform that uses HTML to format a word document. This is useful for organizing the files, and having better UI access to the reports

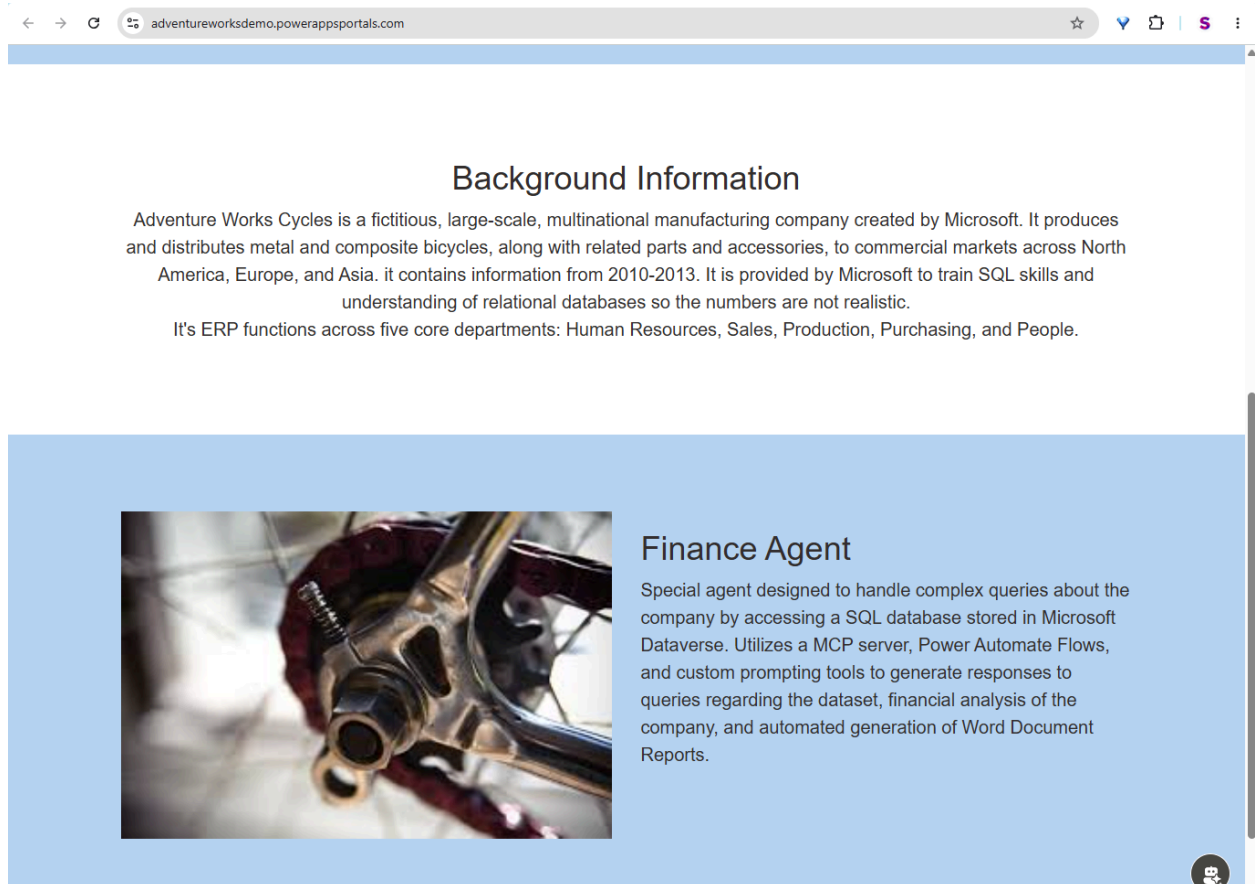


*Various reports that were generated are stored in an easy-to-understand way*

- j. Custom Prompts
  - i. Another tool available in Copilot Studio that allows the addition of special instructions for certain prompts, it was used to specify the details that must be included in certain reports, such as balance sheets and actual vs budget variance reports
- k. Microsoft Power Pages
  - i. A Microsoft Licensed Service that allows the creation of a web application that an iframe from Copilot Studio can be integrated into the website. It was set up to introduce the AdventureWorks company and the agent, while also looking like an ERP system, so that the target user can see the relevant tables and use the agent to ask questions about the data instead of writing sql or manually aggregating each column. The code for this can also be found on GitHub



*Front Page of Adventureworks ERP system*



*Short introduction of the data warehouse and the abilities of the agent*

adventureworksdemo.powerappsportals.com/Dashboard/Internet\_Sales/

Search

| SalesOrderNumber | Created On        | CustomerPONumber | DiscountAmount | CurrencyKey | ExtendedAmount |
|------------------|-------------------|------------------|----------------|-------------|----------------|
| SO43697          | 5/29/2026 3:36 PM |                  | 0.00           | 19          | \$3,578.27     |
| SO43698          | 5/29/2026 3:36 PM |                  | 0.00           | 39          | \$3,399.99     |
| SO43699          | 5/29/2026 3:36 PM |                  | 0.00           | 100         | \$3,399.99     |
| SO43700          | 5/29/2026 3:36 PM |                  | 0.00           | 100         | \$699.10       |
| SO43701          | 5/29/2026 3:36 PM |                  | 0.00           | 6           | \$3,399.99     |
| SO43702          | 5/29/2026 3:36 PM |                  | 0.00           | 100         | \$3,578.27     |
| SO43703          | 5/29/2026 3:36 PM |                  | 0.00           | 6           | \$3,578.27     |
| SO43704          | 5/29/2026 3:36 PM |                  | 0.00           | 6           | \$3,374.99     |
| SO43705          | 5/29/2026 3:36 PM |                  | 0.00           | 6           | \$3,399.99     |
| SO43706          | 5/29/2026 3:36 PM |                  | 0.00           | 100         | \$3,578.27     |

Finance Agent

Connecting...

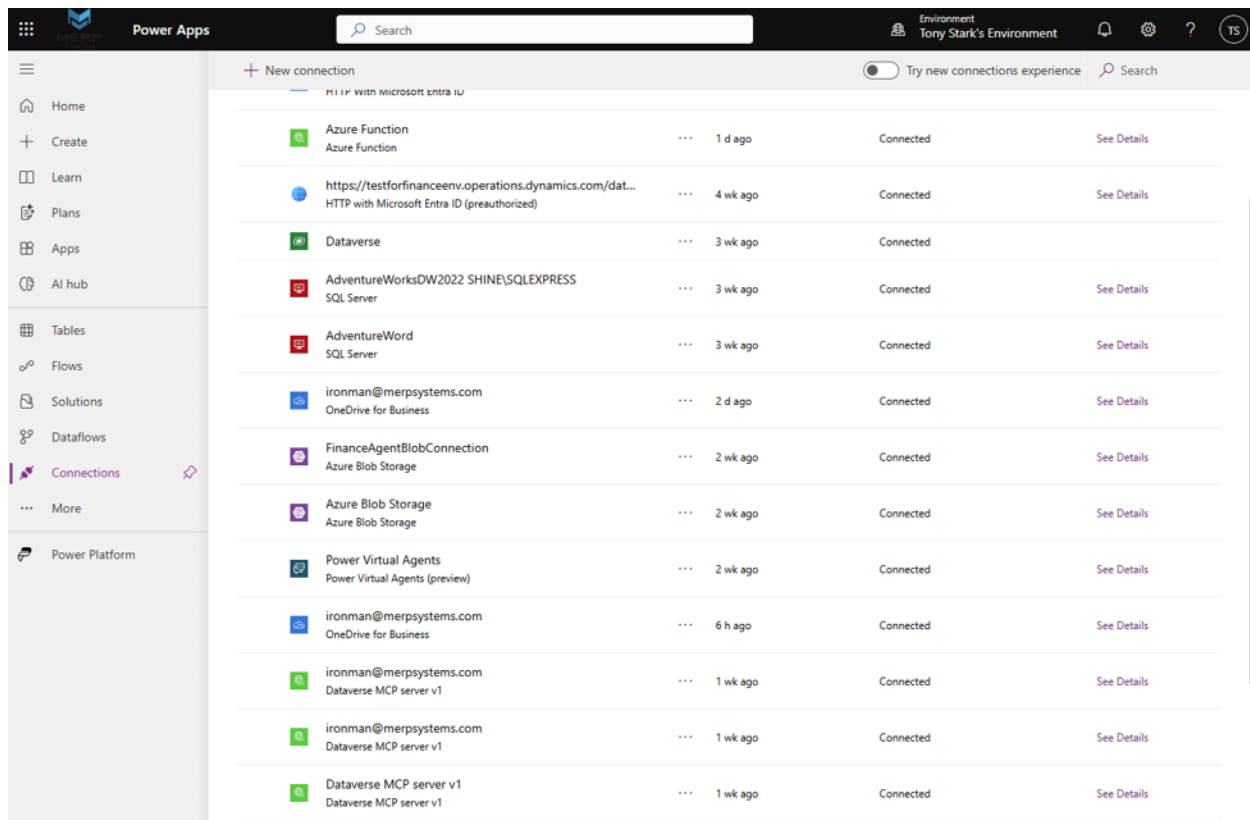
Describe what you'd like help with

Make sure AI-generated content is accurate and appropriate before using.

*Under pages, tables like internet sales, reseller sales, finance details, and product details are available. This is the internet sales page, which has the raw data table as well as an agent that can be used to do any task related to any tables.*

#### I. Power Automate Connectors

- i. In order to connect these services together, this service was used. To avoid OAuth, certain custom connectors were made, while others were remade due to the connectors timing out.



## 12. Security Model

- Agent authenticates to Dataverse using service principal and client credentials in the flow
- No Auth was chosen to accommodate the iframe embedding requirement, and based on the assumption that anyone with access to the ERP system is also a trusted user of the agent
- Azure function handles its authentication independently by relying on Azure storage credentials like client ID, tenant ID, and client secret instead of running user credentials
- A custom security role called AdventureWorks DW Editor was created to allow only all reading data, as this is an old dataset

## 13. Challenges & Solutions

1. Challenge: The Dataverse MCP Server was accessing all tables in Dataverse. Creating an MCP user allowed role filtering to block data in other tables, but the MCP server could still access the metadata. Solution: Using a combination of custom roles and agent instruction as an enforcement layer
2. Challenge: The data type of certain columns was wrong when importing, such as date as a string in FactFinance, resulting in incorrect calculations and queries based on date not processing. Solution: Use CAST for all date joins and ensure all numbers are saved as integers and not strings
3. Challenge: Double counting when making joins due to hierarchy. Both the parent and the child rows were being aggregated, resulting in illogical numbers. Solution: filter based on if the parent organization is null

4. Challenge: The MCP server was unable to join and was guessing relations based on the naming convention of the columns. Solution: Added the database schema as a knowledge source

#### 14. Test Cases & Validation

1. Before adding the Azure Function, the Power Automate flow was tested separately in the test environment to see if formatting and link generation were done properly.
2. The MCP server was tested desperately to ensure it could do tasks such as describe and search. Ex. What were the top 5 customers? How many products were sold on internet sales in Q4 2010, and what departments does the company have?
3. Before deploying the Azure function to Azure, it was tested locally in VS Code, the function URL was confirmed to ensure JSON validation, environment variables were tested using a separate test script called test\_tds.py. The TDS endpoint connectivity was confirmed by testing port and ensuring there are no firewalls in place  
Tested connection isolation Ex. 

```
python  
print("Step 1: Connecting...")  
conn = pyodbc.connect(conn_str, timeout=30, autocommit=True)  
print("Step 1 SUCCESS")  
  
print("Step 2: Creating cursor...")  
cursor = conn.cursor()  
print("Step 2 SUCCESS")  
  
print("Step 3: Executing query...")  
cursor.execute("SELECT 1")  
print("Step 3 SUCCESS:", cursor.fetchone())
```
4. Tested table queries once all background checks were done to test end to end connection of the TDS endpoint, Dataverse, and JSON response ex. 

```
$response = Invoke-WebRequest -Uri "http://localhost:7071/api/query-dataverse" -Method POST -ContentType "application/json" -Body '{"query": "SELECT TOP 5 crfaa_organizationname FROM crfaa_dimorganizationaw"}' -UseBasicParsing
```
5. Tested Multitable joins for example budget vs actual analysis, which resulted in 

```
```json
```
6. 

```
{"columns": ["crfaa_organizationname", "Actual", "Budget"], "rows": [], "count": 0}
```

  
No error was thrown — the query executed successfully but matched no rows. This indicated a silent JOIN failure rather than a syntax or permission problem. To fix had to adjust how dates were saved in Dataverse and which tables were considered leaf nodes
7. Once the Azure function was considered to work properly, testing took place in the Copilot Studio test environment using cases like: budget vs variance reports, profit and loss report, balance sheets, as well as non report queries like product descriptions and customer analysis. Custom prompting and agent instructions were used to enhance accuracy at this stage
8. Once all backend is confirmed to be working, the same tests were conducted in a demo website as well as the iframe in Power Pages.

## 15. Limitations & Known Issues

- The agent's speed is quite slow considering current AI technology.
- It is also not good at prompting the user for additional details when needed.
- It works best when the relational database scheme is known and defined in a knowledge source. It would not work for larger ERP systems that have hundreds of tables.
- AdventureWorks is an old dataset that only has 4 years worth of data. Therefore, this agent was not tested to analyze trends over multiple years or assess the current financial standing of a company
- Prompt injection to the MCP server could be possible, as it does not have the hard control a custom MCP server would

## 16. Conclusion

This project started with broad ideas about the nature of on premise servers, the organization of SQL servers, the usability of ERP systems, and the need to automate repetitive report writing. This project experimented with different ways to store a serverless SQL server, different ways to query a relational database, and how to create meaningful reports out of aggregated data. Explanations were added to provide clarity and avoid the black box agent problem; the agent's orchestration logic was followed to ensure accuracy. Despite struggling with issues like schema and authorization, this experiment helped inform the company I interned at about the possibilities of creating more nuanced agents for larger ERP systems.

## 17. Glossary

- a. Copilot Studio-Microsoft's low-code platform for building AI agents (formerly Power Virtual Agents). Used here to orchestrate tool calls, custom prompts, and natural language responses
- b. Generative orchestrator-The Copilot Studio agent's core mode — the LLM decides which tool to call, what query to build, and how to format the response, based on instructions and the user's intent.
- c. MCP server-Model Context Protocol server. A standard interface that exposes tools and data sources to an AI agent. Used here for the built-in Dataverse MCP providing schema discovery and simple reads.
- d. MCP tool-An individual function exposed by an MCP server — e.g. `list_tables`, `describe_table`, `read_query`. The agent calls these as part of its reasoning loop.
- e. Custom prompt-A named prompt stored in Copilot Studio loaded before tool calls for specific report types. Used here for schema reference, balance sheet rules, and budget vs actual rules.
- f. Agent instructions-The system-level prompt inside the Copilot Studio agent defining tool routing logic, report rules, table allowlist, and output constraints.
- g. Tool routing-The agent's decision of which tool to call for a given query. In this project: MCP for simple single-table reads, Azure Function for aggregations and multi-table joins.

- h. No auth (agent)-Authentication mode on the Copilot Studio agent that allows anonymous access — required for iframe embedding in Power Pages or static HTML. Distinct from the function's own internal auth to Dataverse.
- i. Star schema-A database design where a central fact table is surrounded by dimension tables — like a star. Optimised for analytical queries with simple joins.
- j. Snowflake schema-An extension of star schema where dimension tables are further normalised into sub-dimensions — e.g. DimProduct → DimProductSubcategory → DimProductCategory.
- k. Dataverse-Microsoft's cloud database platform, part of Power Platform. Used here to host the imported AdventureWorks DW tables, accessible via MCP and TDS endpoint.
- l. Dataflow-A Power Platform ETL tool used to import AdventureWorks tables from SQL Server into Dataverse. Similar to Power Query — connects, transforms, and loads data.
- m. Semi-additive measure-A metric that can be summed across some dimensions but not all — e.g. inventory balance can be summed by product but not by time. FactFinance amounts are semi-additive.
- n. Organisation hierarchy-The parent-child structure in DimOrganization. Summing FactFinance across all levels double-counts rolled-up amounts. Filter with WHERE ParentOrganizationKey IS NULL to get top-level only.
- o. Service principal-An application identity in Microsoft Entra ID (Azure AD) used for non-interactive, machine-to-machine authentication. The Azure Function authenticates to Dataverse using a service principal.
- p. Client credentials flow-An OAuth 2.0 grant type where an application authenticates using its own client ID and secret — no user sign-in required. Used by the Azure Function to obtain a Dataverse access token.
- q. App registration-An entry in Microsoft Entra ID representing the application identity. The "Dataverse MCP - Copilot Studio v1" app registration provides the CLIENT\_ID and CLIENT\_SECRET used throughout.
- r. Application user-A non-interactive Dataverse user linked to an app registration. Must be created in Power Platform Admin Center and assigned a security role before the Azure Function can query data.
- s. Security role-A Dataverse permission set controlling table-level access (read, create, write, delete) at organisation, business unit, or user level. The "AdventureWorks DW Editor" role was created for this project.
- t. ActiveDirectoryServicePrincipal-The pyodbc authentication keyword used to connect to the Dataverse TDS endpoint using a service principal (CLIENT\_ID as UID, CLIENT\_SECRET as PWD) instead of a user password.
- u. TDS endpoint-Tabular Data Stream endpoint exposed by Dataverse at port 5558. Accepts real T-SQL queries including JOINS, CASE, GROUP BY, and subqueries — unlike the MCP's read\_query which is more restricted.
- v. pyodbc-A Python library for connecting to databases via ODBC. Used with ODBC Driver 18 for SQL Server to connect to the Dataverse TDS endpoint. Requires autocommit=True for Dataverse compatibility.

- w. ODBC Driver 18-Microsoft's SQL Server ODBC driver required by pyodbc to connect to the TDS endpoint. Must be installed on both local dev machines and Azure deployment environments.
- x. autocommit=True-A pyodbc connection setting required for the Dataverse TDS endpoint. Without it, the connection establishes but silently drops on the first query execution (08S01 communication link failure).
- y. Power Automate-Microsoft's workflow automation platform. Used here to receive query results from the agent, build the HTML report, save to SharePoint, generate a SAS URL, and return the link.
- z. Agent flow-A Power Automate flow triggered directly by a Copilot Studio agent. Runs under the flow maker's credentials — compatible with No Auth agents because the flow's own auth is independent of the agent's auth.
- aa. CAST-A T-SQL function used to convert between data types. Critical in this project for the datekey mismatch fix: CAST(f.crfaa\_datekey AS VARCHAR(8)) aligns the integer fact key with the string dimension key.
- bb. Fan-out (double counting)-A query result inflation caused by joining tables with one-to-many relationships at multiple hierarchy levels simultaneously. Prevented here with the ParentOrganizationKey IS NULL filter on FactFinance queries.
- cc. SAS URL-Shared Access Signature URL — a time-limited link to an Azure Blob Storage resource. Used here to give users a 24-hour download link to their generated report without making the storage container public.
- dd. FactFinance granularity-FactFinance stores amounts at rolled-up account-type level only (Assets, Liabilities, Revenue, Expenditures). Individual GL line items like Cash, Payables, or Salaries do not exist as separate rows.
- ee. ERP-Enterprise Resource Planning — integrated business management software covering finance, inventory, HR, and operations. This project simulates an ERP analytics layer using AdventureWorks DW as the data source.